

Majock Bim

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EDUCATION

McMaster University

Bachelor of Engineering in Mechatronics Engineering

Hamilton, ON

Sept 2025 – May 2029

EXPERIENCE

Embedded & Controls Software Team Lead

June 2026 – Present

McMaster Exoskeleton

Hamilton, ON

- Leading an 8-member embedded software team developing sensing, communication, and motor-control firmware for a lower-limb exoskeleton.

Embedded Software Engineer

Sept 2025 – May 2026

- Optimized STM32 IMU acquisition in C by increasing the I²C bus clock from 100 to 400 kHz and implementing DMA-based reads with interrupt-driven completion.
- Developed CAN communication firmware linking a Raspberry Pi 5 and four STM32 controllers over a 1 Mbit/s bus for IMU telemetry and joint torque commands.

PROJECTS

Physio (LA Hacks 2026) | *Embedded C++, ESP32, Edge ML, iOS*

- Built a stroke-rehabilitation wearable around an ESP32-C3 and two MPU6050 IMUs, resolving shared-bus address conflicts and adding an OLED interface for hardware debugging.
- Implemented ESP32-to-iOS BLE telemetry, streaming 76-byte kinematic packets at 80 Hz to an iOS device.
- Designed a two-layer carrier PCB in Altium Designer for a compact rehabilitation wearable; generated Gerbers, BOM, and pick-and-place files, earning a PCBWay manufacturing sponsorship for a solo second iteration.
- Co-deployed a PyTorch 1D CNN to the iOS app using Zetic MLange for movement scoring, with platform-reported inference latency as low as 0.04 ms.

Spectrum - Terminal Audio Visualizer | *C++17, FFTW3, WASAPI*

- Developed an open-source C++17 terminal spectrum analyzer utilizing WASAPI loopback and FFTW3, achieving < 5% CPU usage and a consistent 60 FPS render loop (100+ downloads, 40+ GitHub stars).
- Implemented a DSP pipeline using FFTW3 with 2,400-sample Hann windows, 120-sample hops, and automatic gain control for real-time audio visualization.
- Built a multithreaded audio pipeline separating audio capture and FFT processing from terminal rendering, using sample accumulation and mutex-protected buffers to share results.

VerifAI - Media Verification Platform | *Python, FastAPI, OpenCV, NumPy, Docker*

- Built a Python/FastAPI image-forensics pipeline using OpenCV and NumPy for PCA, FFT analysis, and Error Level Analysis to assess potential media manipulation.
- Integrated Gemini to synthesize forensic signals into structured verdicts with supporting evidence; containerized and deployed the backend on DigitalOcean to offload processing from clients.

ADDITIONAL EXPERIENCE

Developer & Code Evaluator

May 2026 – Aug 2026

OutlierAI, DataAnnotation

Remote

- Evaluated and debugged C++ and Python code snippets, focusing on memory management and algorithmic efficiency, to improve the accuracy of foundational LLM models.

TECHNICAL SKILLS

Languages: C++, C, Python, Go, MATLAB

Embedded & Systems: STM32, Raspberry Pi 5, ESP32, BLE, Arduino, CAN bus, I²C, UART, DMA, IMU, Linux

Frameworks & Tools: Git, Vim, Altium Designer, STM32CubeIDE, OpenCV, Google Cloud